

Unit 2 – Exploring Two Variable Data

Notes 1 – Two Categorical Variables

In Unit 1, we explored how bar graphs are a commonly used way to display categorical data. When we have two categorical variables to compare, we still use bar graphs, but they become a segmented bar graph or a mosaic plot.

Here is our data from Unit 1 HW 1 on the breakdown of state locations and how the state voted in the 2020 election:

		Region				Total
		Northeast	Midwest	South	West	
Party	Democrat	9	4	4	8	25
	Republican	0	8	12	5	25
	Total	9	12	16	13	50

A **side by side bar graph** merges two bar graphs into one, in an attempt to compare the distributions of the two categorical variables.

A **segmented bar graph** is another way to display data, where each group is split up by its relative frequency.

A **mosaic plot** is similar to a segmented bar graph, but it draws attention to the sizes of each group.

When you are working with two categorical variables, most of the time, the raw data is given in the form of a two way table - it is a table listing two categorical variables whose values have been paired. Each set of numbers in a two-way table has a specific name.

Joint relative frequencies: ratio of the frequency in a cell and the total number of data values.

Marginal relative frequencies: ratio of the sum of in a row or column and the total number of data values.

Conditional relative frequencies: ratio of a joint relative frequency and related marginal relative frequency.

- This would be when you want to find the percentage of people that _____, given _____.
- This is a similar set up to conditional probability, where the condition, is preceded by the word given. This will be covered in more detail in Unit 4.

Ex)

		Region				Total
		Northeast	Midwest	South	West	
Party	Democrat	9	4	4	8	25
	Republican	0	8	12	5	25
	Total	9	12	16	13	50

Joint Relative Frequency	Marginal Relative Frequency	Conditional Relative Frequency
Percent of states that are in the Midwest and voted "Democrat"	Percent of states that are in the South.	Of the states that voted "democrat", the percent that is from the West.

Ex) The director of a technical school was curious about whether there is a relationship between students who complete one of the school's most popular health sciences certificate programs and whether those students go on to complete more advanced studies in the health sciences within two years of completing the certificate program. She randomly selected 100 students who completed the program. Data collected on these students are shown in the table below.

		Completed more Advanced Studies		Total
		Yes	No	
Completed Most Popular Health Science Certificate Program	Yes	35	25	60
	No	5	35	40
	Total	40	60	100

Which statement is true about these 100 students? (show work proving/disproving/explaining)

(A) Being a person who completed more advanced studies is more likely than being a person who did not complete more advanced studies.

(B) Being a person who completed program is less likely than being a person who did not complete the program.

(C) Being a person who completed the program and completed more advanced studies is less likely than being a person who did not complete the program and did not complete more advanced studies.

(D) Being a person who did not complete the program but completed more advanced studies is less likely than being a person who completed the program and completed more advanced studies.

(E) Being a person who completed the program but did not complete more advanced studies is more likely than being a person who did not complete the program and did not complete more advanced studies.

Ex) A 1969 study among the Pima Indians of Arizona investigated the relationship between a mother's diabetic status and the appearance of birth defects in her children. The results appear in the two-way table below.

		Diabetic Status			Total
		Nondiabetic	Prediabetic	Diabetic	
Birth Defects	No	754	362	38	
	Yes	31	13	9	
	Total				

(a) Fill in the row and column totals in the margins of the table.

(b) Compute (in percent) the conditional distributions of birth defects for each diabetic status.

(c) Display the conditional distributions in a segmented bar graph. Don't forget to label your graph completely. Comment on any clear associations you see.

Ex) Fourth grade children were asked what emotion they associated with the color red. The responses for emotion and gender of the children are summarized in the following two-way table.

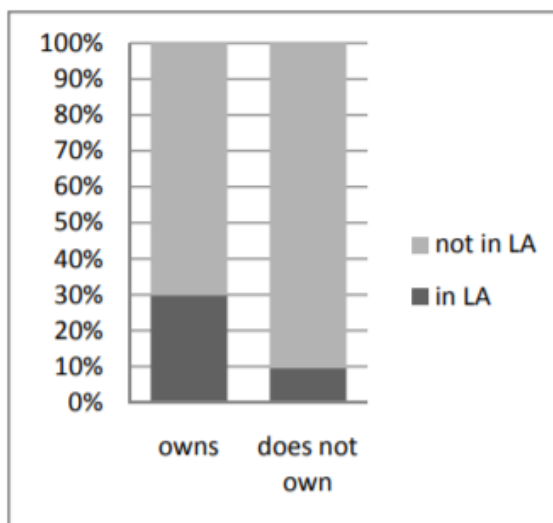
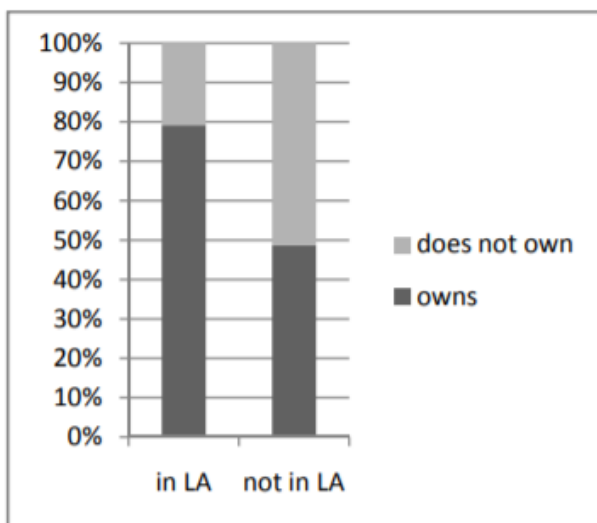
	Anger	Pain	Happiness	Love
Male	35	27	12	38
Female	27	17	19	39

What proportion of the males associate the color red with love?

- (A) 0.5234
- (B) 0.3598
- (C) 0.3393
- (D) 0.1822
- (E) 0.1775

Ex) There's a stereotype in that everybody in LA has a cell phone. A researcher decided to collect data to determine if LA people were indeed more likely to own cell phones than other Californians. The two-way table and two different bar graphs were created.

	From LA	From somewhere else in CA	Total
Owns a cell phone	164	386	550
Does not own a cell phone	43	407	450
Total	207	793	1000



a) Which plot makes it easier to answer the question “Are LA people more likely to own a cell phone than other Californians?” Why?

b) What does the other segmented bar graph tell you? Write a sentence or two based on the other graph.